Short Guide: What is a CDN?

What it is

A Content Delivery Network (CDN) is a group of servers spread across many locations. It stores copies of static content such as images, CSS, JavaScript and PDF files, and serves them from the location closest to the visitor. This reduces loading time and lowers the load on the original server.

Key terms

- **Origin:** the server that holds the original files (here, GitHub Pages).
- **Edge / POP:** a CDN server close to users that keeps cached copies.
- **Cache hit:** the edge already has the file and returns it directly.
- **Cache miss:** the edge fetches the file from the origin, then stores it.
- **TTL:** how long a cached copy stays valid before it is rechecked.

Why different TTL values

Files that rarely change, like images, can be cached for days. Data that changes often, like a news JSON file, needs a very short TTL so visitors do not see outdated content. When content must update immediately, an administrator can purge (refresh) the cached URL.

This file is part of a CDN homework project using GitHub Pages as the origin.